Pradeep Singh

Bachelor of Technology
in Computer Science Engineering
Indian Institute Of Technology, Ropar

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GitHub | Website
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Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	8.55 (Till 6th Sem)	2023–27
Senior Secondary	Central Board of Secondary Education	90%	2022–23
Secondary	Central Board of Secondary Education	89.8%	2019–20

Experience

- **Pattern Technologies India Private Limited** *Jun–Jul 2026*
  Software Engineer Intern
  – Developed a full-stack web application for payroll automation, integrating Snowflake for data pipeline workflows and consuming Snowflake data using the Node.js Snowflake SDK.
  – Integrated LLM-powered APIs through Bifrost for structured data extraction from sensitive documents, while applying data security practices including AWS KMS encryption.
  – Gained an understanding of Pattern's data architecture and workflows, including how data is collected, processed, stored, and consumed across production systems.

Major Projects

- **Snehita Well-Being (Mental Health Counselling Platform, IIT Ropar)** *Feb–May 2026*
  Full-Stack Web Application **GitHub**
  – Built a secure full-stack platform for mental health appointment scheduling and management.
  – Developed RESTful APIs with JWT authentication and role-based access control for secure user and counsellor workflows.
  – Designed a responsive React frontend for appointment booking, user profiles, and availability management.
  – Architected an MVC-style backend using Node.js and Express.js with Sequelize ORM and PostgreSQL, and containerized the application using Docker.
- **RISC-V Assembler with 5-Stage Pipeline Simulator** *Feb–Apr 2025*
  Computer Architecture **GitHub**
  – Built an assembler and a 5-stage pipelined CPU simulator implementing most RISC-V instructions, following the RISC-V ISA.
  – Designed a Branch Target Buffer (BTB) with 1-bit prediction to reduce control hazards.
  – Implemented both pipelined and non-pipelined execution, with configurable toggles for data forwarding and branch prediction.

Minor Projects

- **AIMS – Academic Information Management System** *Jan–Feb 2026*
  Full-Stack Role-Based Web Application **GitHub**
  – Developed a role-based academic management system supporting Student, Instructor, Advisor, and Admin workflows.
  – Implemented multi-level course enrollment approval (Instructor → Faculty Advisor) reflecting real university processes.
  – Built RESTful APIs with Express.js and integrated Supabase (PostgreSQL).
- **Cache Performance in Traffic Simulation: LRU vs Splay Tree** *June 2025*
  DSA + Systems Simulation **GitHub**
  – Implemented Dijkstra's algorithm on a city graph with LRU and Splay Tree caches to avoid redundant path computations.
  – Benchmarked cache hit/miss rates under dynamic traffic conditions.
  – Highlighted performance benefits of self-adjusting data structures in real-time systems.

Technical Skills

- **Programming Languages**: C/C++, Python, JavaScript
- **Web Development**: React.js, Node.js, Express.js, HTML, CSS
- **Databases & Tools**: PostgreSQL, MySQL, Git, GitHub, Linux, Docker
- **Concepts**: Data Structures and Algorithms, OOP, Computer Networks, Operating Systems

Key courses taken

- **CSE & Maths**: Machine Learning ,Computer Networks, Software Engineering, Operating System, Database Management System, Algorithms & Data Structures, Computer Architecture,Financial Risk Management,Linear Algebra, Probability & Statistics
- **Others**: Signal and System, Basic Electronics, Economics

Achievements

- **Codeforces: Specialist (Rating 1499)** – profile – 4 times in top 10 percent.
- **500+ DSA Problems Solved** – Solved 500+ problems across **LeetCode** and **GeeksforGeeks**.
- **Inter-IIT Sports Meet** Participant and intra-college sports Meet Silver Medalist (2024).